

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter 2022

Course: B. Tech

Subject: Engineering Chemistry

Marks: 60

Date: 23.03.23

Sem: I

Subject code: BTBS102

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Draw a neat diagram wherever necessary
3. Figures to right indicates full marks

		(Level /CO)	Marks
Q.1 Solve Any TWO of the following.			
A)	Explain determination of Dissolved Oxygen (DO) in water by Winkler's/Iodometric method.	2	06
B)	Discuss disadvantages of hard water in Domestic and Industrial use.	1	06
C)	Explain in detail Hot Lime-Soda process of softening of water with its advantages and disadvantages.	2	06
Q.2 Solve Any TWO of the following.			
A)	Explain the Phase diagram of one component Water system.	2	06
B)	Discuss the term Phase & Degree of freedom involved in Phase Rule equation with examples.	1	06
C)	Explain Phase diagram of two component Ag-Pb alloy system.	2	06
Q.3 Solve Any TWO of the following.			
A)	Describe in brief Direct (Dry) chemical corrosion.	1	06
B)	Discuss Hydrogen evolution mechanism involved in electrochemical corrosion.	2	06
C)	Explain any one method to minimize the rate of corrosion.	2	06
Q.4 Solve Any TWO of the following.			
A)	What is Coal? Explain various types of coal.	1	06
B)	Explain Proximate analysis of coal.	2	06
C)	Describe in brief solid, semi-solid & liquid lubricants.	2	06
Q.5 Solve Any TWO of the following.			
A)	Write a note on conductometric titrations with suitable examples.	1	06
B)	Describe Ostwald's theory of Acid-Base indicator.	2	06
C)	Discuss H ₂ -O ₂ Fuel cell with its advantages.	2	06

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